

Philip (Phi) Bui

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EDUCATION & HONORS

- Ph.D. in Urban Systems, New York University, New York, NY** Sep 2026 - Present
- Researching under Dr. Debra Laefer in the Urban Modeling Group; Awarded Tandon's School of Engineering Fellowship.
- M.S. in Computer Science: Human-Robot Interaction, Tufts University, Medford, MA** Sep 2024 - May 2025
- Graduated with a 3.8/4.0 GPA; Inducted into the Honos Civicus Society (Jonathan M. Tisch College of Civic Life) for civic impact through Untitled LLC.
- B.S. in Computer Science, Worcester Polytechnic Institute (WPI), Worcester, MA** Aug 2021 - Dec 2023
- Graduated with a 3.8/4.0 GPA and distinction; Dean's List; Upsilon Pi Epsilon (Computer Science Honor Society).

HIGHLIGHTS

- AI Platform for Low-Income Housing Development for City of Worcester, MA, WPI MQP** Aug - Dec 2023
- Partnered with Worcester's Chief Development Officer to build an AI-assisted land-acquisition and financing pipeline. Integrated GIS layers and open municipal data into a PostgreSQL/Node.js backend and React dashboard.
 - Surfaced parcel-level transparency for civic stakeholders. Published records onto blockchain for transparency and accountability. Now integrated with the City of Worcester's low income housing development pipeline.
- State-Wise Constrained Policy Shaping: Runtime Behavior Steering for Safe RL, Tufts University** Jan 2024 - 2025
- With Dr. Vasanth Sarathy. Presented @ AAAI-26: AIGOV in Singapore.
 - Designed a post-training supervisor enforcing safety norms on DQN agents in HighwayEnv; Reduced collisions by 97% in-distribution and 99% zero-shot.
- Graduate Research Assistant, Human Interaction Lab, Tufts University, Medford, MA** Jan - May 2025
- With Dr. Saul Albert. Presented @ ICCA-26 at University of Alberta.
 - Led the back-end overhaul of **GailBot**, a PyPI-distributed transcription pipeline for Jeffersonian conversation-analytic notation, built atop off-the-shelf speech-to-text (DeepSeek, Watson, Google, OpenAI)
 - Engineered REST APIs, designed the PostgreSQL schema, implemented full CRUD in React; authored cross-platform deployment docs and resolved critical Linux/macOS regressions.

RESEARCH EXPERIENCE

- Research Intern, AI Futures Collab, WPI, Worcester, MA** May 2025 - Sep 2025
- With Dr. Yunus Telli. Investigating human/AI interfacing and literacy, evaluating phenomenology, affordances, AI interactions, transparency and ethical implications of AI-driven media.
- Swarm Robotics AI Research Intern, Air Force Research Laboratory, Eglin Air Force Base, FL** May - Aug 2025
- With Dr. Scott Nivison. Designed heterogeneous multi-agent reinforcement-learning algorithms for swarm coordination and decentralized decision-making under communication constraints. (Classified; details omitted.)
- LLM Enhanced Captioning for Neurodivergent Users, Tufts University** Jan - May 2025
- Co-developed a real-time Chromium extension integrating a quantized Mistral-7B model over WebSocket; enriched captions with sentiment and intent, achieved <200 ms latency & >95% word accuracy; two pilots (N=8, N=3) reported 4.5/5 clarity.
- LLMs Can Be Judgy Too: Fine-Tuning LLMs for Character-Driven RL, Tufts University** Dec 2024 - May 2025
- Built a PPO pipeline using a fine-tuned LLM as a reward model for facial-AU generation; across 30 scenarios, LLM-only achieved 3.12/5 mean accuracy, the hybrid approach matched at 3.07/5 with 32% less variance, and halved manual labeling effort.

Computer Vision-Guided Sphero Swarm Navigation Framework, Tufts University

Sep - Dec 2024

- Integrated overhead-camera MCL, PRM path planning, and a BLE-based microservice layer to enable scalable, real-time collision-free coordination of Sphero Bolt swarms.

Evaluating Preferences in Robot Control Systems, Tufts University

Sep - Dec 2024

- Ran an HRI study (n=11) comparing manual stick control vs. autonomous MCL/PRM navigation on obstacle courses; manual was 30% faster with slightly fewer collisions and preferred by 64% of participants.

PROFESSIONAL EXPERIENCE**Full-Stack Software Engineer, Cyvl, Somerville, MA**

Sep 2025 - Present

- Designed the AI ingestion pipeline for municipal engineering documents and diagrams into geometrically-accurate geospatial files; Winning submission to NYC DEP's ETL Challenge; converted into multi-million-dollar contracts.
- Built a visual embedding pipeline indexing 500k+ high-resolution street-level images per city into a Qdrant vector store, enabling sub-second semantic retrieval over urban infrastructure imagery.
- Extended production AI pipelines processing petabyte-scale, georeferenced imagery and LiDAR point clouds to identify and analyze pavement distress, signs, markings, and above-ground assets across municipal road networks.
- Shipped dozens of features adopted by hundreds of executive-level stakeholders across municipal public-works and transportation departments.
- Partnered with City of Boston leadership to build and maintain Model Context Protocol (MCP) servers, giving non-technical civic staff natural-language access to infrastructure datasets and narrowing the data-access gap between city hall and the agencies that serve residents.

Software Engineer, enLabel, Boston, MA

Jan 2024 - Jan 2025

- Refactored hundreds of defects in legacy C#/ASP .NET code; led React/TypeScript UI redesign, improving workflows and user satisfaction.
- Architected a modular React frontend with optimized ASP .NET Web API integration; authored end-to-end regression-testing suite to automate API validation.
- Contributed to technical planning and architecture meetings with cross-functional stakeholders
- Led design specification, timeline management, and strategic coordination of engineering deliverables.

Website Developer, ProsperOn, Boston, MA

Jun 2023 - Feb 2024

- Built accessible React/TypeScript financial-simulation pages from Figma; refactored Node.js RESTful APIs to modernize legacy code.
- Piloted platform integrations with MassMutual, Mass Fintech Hub, and Boston University; featured in Fintech Sandbox as a leading financial-wellness startup.

Operations Intern, Amazon, Salem, OR

Jun - Aug 2023

- Developed a Node.js mapping tool to track >7,000 items per warehouse; integrated into onboarding training, cutting per-employee costs by 95%.
- Served as deputized loss-prevention officer and led inter-shift morale initiatives.

COMMUNITY SERVICE**Community Team Coordinator, Cyvl, Somerville, MA**

Sep 2025 - Present

- Developed system of community outreach to institutions, university organizations, and data communities such as: Yale (PITSCY), NYU (CUSP), Tufts (JumboHacks), BU (Spark), NU (SUN Lab), WPI (UPE), Olin, BetaNYC (NYC School of Data).
- Collaborated with university organizations to bring lectures, workshops, sponsorships, hackathons, and research opportunities to students.
- Developed and maintained tools (APIs, MCPs, agentic tooling schemas) for open data access for cities, institutions, and constituents.

Co-Founder & Chief Logistics Officer, Untitled LLC, Boston, MA

Dec 2023 - Present

- Launched an underground arts collective securing venues, sponsorships, and studio space; produced sold-out fashion shows and performances for thousands of attendees and millions of social-media views.
- Recruited/managed cross-functional teams, handled legal/financial strategy, coordinated security & harm-reduction, and led fundraising for underrepresented artists.
- Inducted into Honos Civicus Society by the Jonathan M. Tisch College of Civic Life at Tufts for Untitled LLC's civic impact in the community.

Front End Developer & Educator, Freelance, Boston, MA

Feb 2023 - Present

- Collaborated with local artists to translate creative visions into responsive digital experiences, and educating them on front-end fundamentals.

Community Events Coordinator, WPI Powerlifting, Worcester, MA

Jan 2023 - Dec 2023

- Forged partnerships with collegiate and professional powerlifting organizations; designed inclusive skill-building workshops, boosting recruitment and diversity.

Instructor, YWCA Central Massachusetts, Worcester, MA

Aug 2021 - Jun 2022

- Delivered swim & fitness instruction to 6–65 year-olds, including students with disabilities; achieved 100% course completion and performed three lifesaving interventions.